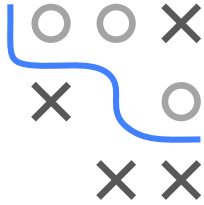
# RISK MINIMIZER AND OPTIMAL CONSTANT

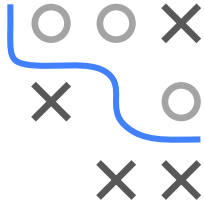
| Name | Risk Minimizer | Optimal Constant |
|---|---|---|
| L2 | $f^*(\mathbf{x}) = \mathbb{E}_{y\|x}[y \mid \mathbf{x}]$ | $\hat{f}(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} y^{(i)}$ |
| L1 | $f^*(\mathbf{x}) = \text{med}_{y\|x}[y \mid \mathbf{x}]$ | $\hat{f}(\mathbf{x}) = \text{med}(y^{(i)})$ |
| 0-1 | $h^*(\mathbf{x}) = \arg\max_{l\in\mathcal{Y}} \mathbb{P}(y = l \mid \mathbf{x})$ | $\hat{h}(\mathbf{x}) = \text{mode}\left\{y^{(i)}\right\}$ |
| Brier | $\pi^*(\mathbf{x}) = \mathbb{P}(y = 1 \mid \mathbf{x})$ | $\hat{\pi}(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} y^{(i)}$ |
| Bernoulli (on probs) | $\pi^*(\mathbf{x}) = \mathbb{P}(y = 1 \mid \mathbf{x})$ | $\hat{\pi}(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} y^{(i)}$ |
| Bernoulli (on scores) | $f^*(\mathbf{x}) = \log\left(\frac{\mathbb{P}(y=1 \mid \mathbf{x})}{1-\mathbb{P}(y=1 \mid \mathbf{x})}\right)$ | $\hat{f}(\mathbf{x}) = \log\frac{n_{+1}}{n_{-1}}$ |



We see: For regression, the RMs model the conditional expectation and median of the underlying distribution. This makes intuitive sense, depending on your concept of how to best estimate central location / how robust this location should be.

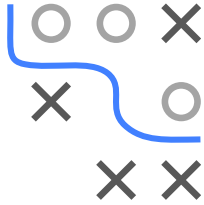
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For the 0-1 loss, the risk minimizer constructs the **optimal Bayes decision rule**: We predict the class with maximal posterior probability.

# RISK MINIMIZER AND OPTIMAL CONSTANT



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For Brier and Bernoulli, we predict the posterior probabilities (of the true DGP!). Losses that have this desirable property are called **proper scoring (rules)**.